

Introduction to Supervised Classification

Confusion Matrix, Accuracy, Precision, Recall, F1-score

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Outline

- 1 Metrics in Machine Learning
 - Mathematical perspective
 - Supervised metrics
 - Unsupervised metrics
- 2 Context: Supervised classification
- 3 Confusion Matrix
- 4 Accuracy
- 5 Precision and Recall
- 6 F1-score
- 7 Beyond binary classification
- 8 Summary

Why are metrics essential?

- A metric = a **mathematical function** that measures the quality of a model.
- They guide:
 - model selection,
 - hyperparameter tuning,
 - stopping criteria,
 - final evaluation.

Key idea

No solid ML/DL without solid metrics. A metric defines what “success” means for a model.

Mathematical perspective

A metric is a function:

$$M : (y_{\text{true}}, y_{\text{pred}}) \mapsto \mathbb{R}$$

which should be:

- **consistent**: close predictions give close scores;
- **interpretable**: bounded or normalized when possible;
- **optimizable**: compatible with learning algorithms.

Example

In classification:

$$\text{Accuracy} = \frac{1}{n} \sum_{i=1}^n 1(y_i = \hat{y}_i)$$

- **Regression**

- MAE:

$$\text{MAE} = \frac{1}{n} \sum |y_i - \hat{y}_i|$$

- MSE:

$$\text{MSE} = \frac{1}{n} \sum (y_i - \hat{y}_i)^2$$

- R^2 :

$$R^2 = 1 - \frac{\sum (y_i - \hat{y}_i)^2}{\sum (y_i - \bar{y})^2}$$

- **Classification (see next slides)**

- Accuracy
- Precision, Recall
- F1-score
- AUC-ROC

When should you choose which metric?

- Balanced classes → **Accuracy**
- False positives are costly → **Precision**
- False negatives are critical → **Recall**
- Strong imbalance → **F1** or **AUC-ROC**

Key message

A metric = a ****business decision****, not just a mathematical one.

Unsupervised learning: no ground truth!

- No y_{true} $\rightarrow$ metrics measure **internal consistency**.
- They quantify:
 - cluster compactness,
 - cluster separation,
 - local density,
 - structural stability.

- **Silhouette score:**

$$s(i) = \frac{b(i) - a(i)}{\max(a(i), b(i))}$$

- $a(i)$ = intra-cluster distance
- $b(i)$ = distance to nearest cluster
- **Davies–Bouldin index** Lower is better.
- **Calinski–Harabasz** Higher is better.

Intuition

A good cluster is both **compact** and **well-separated** from others.

- Distance-based:

$$D(x) = \|x - \mu\|$$

- Density-based (LOF):

$$\text{LOF}(x) = \frac{\text{average local density}}{\text{local density of } x}$$

Practical use

An anomaly = a low-density or isolated point. The metric quantifies this deviation.

Supervised classification: General idea

- Labeled data: **features** + **target label**.
- Goal: learn a function $f(x) \rightarrow y$ that predicts the correct class.
- Examples:
 - Medical diagnosis: sick / healthy.
 - Email: spam / not spam.
 - Product quality: conform / defective.

Key idea

In classification, we do not only care about the **overall error rate**, but about **how** the model makes mistakes.

Confusion matrix: Binary case

- Binary problem: **Positive / Negative**.
- The confusion matrix compares:
 - **True labels**,
 - with **Model predictions**.
- **TP** : True Positive
- **TN** : True Negative
- **FP** : False Positive (“false alarms”)
- **FN** : False Negative (“missed detections”)

	Predicted	
	Positive	Negative
True Positive	TP	FN
True Negative	FP	TN

Simple example

Suppose 100 examples, with the confusion matrix:

	Predicted	
	Positive	Negative
True Positive	40	10
True Negative	5	45

- $TP = 40$, $FN = 10$, $FP = 5$, $TN = 45$.
- Total = 100.

Exercise

Using this matrix, compute: Accuracy, Precision, Recall, F1-score.

Definition

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

- Proportion of **correct** predictions.
- Useful when:
 - Classes are **balanced**.
 - Costs of FP/FN are similar.

Limitation

With **highly imbalanced** data (e.g. 99% negatives), a trivial model can have high accuracy but be useless.

Definition

$$\text{Precision} = \frac{TP}{TP + FP}$$

- Among predicted positives, how many are correct?
- Measures the quality of predicted positives.

When to prioritize it?

When **false positives** are costly. Examples: unnecessary alerts, marketing to uninterested customers.

Recall (Sensitivity)

Definition

$$\text{Recall} = \frac{TP}{TP + FN}$$

- Among true positives, how many are detected?
- Measures the ability to **not miss** positives.

When to prioritize it?

When **false negatives** are critical. Examples: undetected cancer, missed fraud.

F1-score: Precision/Recall trade-off

Definition

$$F1 = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

- Harmonic mean of precision and recall.
- High only if **both** are high.

Interest

- Useful for **imbalanced** datasets.
- Good global indicator of positive-class detection.

Example calculation (with previous matrix)

Recall:

$$TP = 40, \quad FN = 10, \quad FP = 5, \quad TN = 45.$$

- Accuracy:

$$Acc = \frac{40 + 45}{100} = 0.85$$

- Precision:

$$Prec = \frac{40}{40 + 5} \approx 0.89$$

- Recall:

$$Rec = \frac{40}{50} = 0.8$$

- F1:

$$F1 = 2 \times \frac{0.89 \times 0.8}{0.89 + 0.8} \approx 0.84$$

Exercise

Ask students to recompute everything in the notebook

Multi-class case (3+ classes)

- Confusion matrix becomes a $K \times K$ **grid**.
- Precision, recall, F1 can be computed:
 - **Per class** (one-vs-rest),
 - Then averaged:
 - Macro-average (simple mean),
 - Weighted-average (weighted by class size).

Connection to the lab

On datasets like *wine* or *digits*, ask students to display the confusion matrix and per-class F1.

Summary of metrics

- **Confusion matrix:** foundation for all analysis.
- **Accuracy:** global view, beware class imbalance.
- **Precision:** quality of predicted positives.
- **Recall:** ability to detect positives.
- **F1-score:** precision/recall compromise.

Key message for the lab

Always choose metrics based on **business context**: cost of FP/FN, class imbalance, criticality, etc.